

DRAFTGPT

AI-Powered Manufacturing Detailing Platform

From 3D CAD model to shop-floor-ready manufacturing documentation

Product Concept Document

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The core idea: We are not building a simple 3D-to-2D converter. We are building an engineering automation system that understands a 3D part, determines how it needs to be documented, and produces a detailed drawing that can move through engineering review toward manufacturing.

1. Executive Summary

DraftGPT is an AI-powered engineering platform designed to transform a 3D CAD model into a complete, company-standard, manufacturing-ready detailing sheet. The objective is not merely to generate front, top, and side projections. The system must understand the geometry and manufacturing-relevant features of the part, decide how those features should be communicated, generate dimensions and annotations, apply drawing standards, validate the result, and present it for engineer review.

The final output should contain the information required by a machinist, fabricator, tool-room engineer, inspection team, or manufacturing vendor to understand what must be made and how the component should be controlled.

Core product promise: Automate the journey from 3D design to manufacturing-ready documentation.

2. The Problem We Are Solving

A typical engineering workflow requires significant manual drafting and detailing work:

- Open or receive the 3D CAD model.
- Create a drawing and select appropriate views.
- Create sections and detail views when necessary.
- Identify important manufacturing features.
- Add overall and feature-level dimensions.
- Add hole callouts, hole tables, tolerances and annotations.
- Add material, manufacturing and inspection notes.
- Apply company-specific templates and drafting standards.
- Check the drawing for completeness and readability.
- Revise the drawing and release it to manufacturing.

The repetitive nature of this work becomes expensive when an organization creates hundreds or thousands of drawings. The real problem is therefore not simply converting geometry from 3D to 2D; it is automating the engineering decisions required to make a drawing useful.

3. What We Are Building — and What We Are Not

Simple 3D-to-2D converter	DraftGPT
Creates geometric projections	Creates manufacturing documentation
Shows what the part looks like	Communicates what needs to be manufactured
Produces basic views	Plans views, sections and details
May show geometry only	Adds dimensions, annotations, notes and tables

Simple 3D-to-2D converter	DraftGPT
Generic output	Applies company drawing standards
No engineering validation	Runs drawing-quality and confidence checks
End at exported drawing	Includes engineer review and approval workflow

4. What Does “Shop-Floor Ready” Mean?

A drawing is shop-floor ready when the person manufacturing the component has enough reliable information to manufacture and inspect it without repeatedly going back to the design engineer for basic clarification.

Typical drawing information can include:

- Geometry: front, top, side, isometric, section and detail views.
- Dimensions: overall dimensions, feature sizes, hole locations, diameters, radii, depths, angles and thicknesses.
- Dimension strategies: baseline, chain, ordinate and coordinate dimensions where appropriate.
- Feature documentation: hole callouts, hole patterns, hole tables, slots, pockets, chamfers and fillets.
- Engineering controls: tolerances, datums and GD&T; where required.
- Manufacturing information: material, finish, heat treatment, coating, welding and machining notes where applicable.
- Documentation: part number, drawing number, revision, quantity, title block and revision history.

Not every drawing needs every item. DraftGPT must determine what is relevant to the specific part and the company's standards rather than blindly adding annotations.

5. Product Architecture

The system should be built as a pipeline of specialized engineering and AI components:

Layer	Responsibility
1. Model Ingestion	Accept STEP, IGES and supported CAD inputs; preserve source and metadata.
2. CAD Geometry Engine	Read exact topology, surfaces, edges, solids and measurable geometry.
3. Feature Recognition	Identify holes, cylinders, planes, pockets, slots, patterns, chamfers, fillets and other manufacturing-relevant features.
4. Engineering / Rules Engine	Apply deterministic dimensioning, view and drafting rules.
5. AI Reasoning Layer	Assist with interpretation, drawing strategy and ambiguous cases without inventing geometry.
6. Drawing Engine	Create views, sections, dimensions, annotations, tables and notes.
7. Template Engine	Apply company-specific title blocks, layers, standards, units and visual conventions.

Layer	Responsibility
8. Validation Engine	Check geometry coverage, dimension completeness, conflicts, layout and missing information.
9. Review Workflow	Present generated drawing and flagged items to an engineer for correction and approval.
10. Export / Release	Produce DWG, DXF, PDF and other required outputs.

6. Understanding the 3D Model

The CAD geometry engine provides the mathematical ground truth. The platform must not rely on an LLM guessing dimensions from an image when exact geometry is available.

Example feature interpretation:

Cylinder → hole candidate → through/blind classification → diameter → depth → location → pattern relationship.

Multiple identical holes → pattern detection → spacing and location analysis → hole callout or hole table.

Two parallel planar faces → thickness feature → measurable thickness dimension.

This layer converts raw CAD topology into a structured engineering representation that the rest of the system can use.

7. AI + Deterministic Engineering Logic

AI should be an important component, but it should not be the source of geometric truth. The platform should combine exact CAD computation, deterministic engineering rules and AI reasoning.

CAD engine: establishes exact geometry and measurements.

Rules engine: applies predictable engineering and drafting logic.

AI layer: helps interpret complex situations, select strategies and prioritize review.

Principle: the AI should never invent a dimension that can be calculated from the model. Where confidence is low, the system should flag the item for engineer review.

8. End-to-End Workflow

1. 3D CAD model uploaded or selected.
2. Model ingestion validates and stores the source.
3. CAD geometry is parsed into exact topology and measurable entities.
4. Manufacturing-relevant features are recognized and classified.
5. Engineering rules and AI determine the required drawing views.
6. The system plans dimensions and annotation strategies.
7. Hole callouts, hole tables, notes, tolerances and other required documentation are generated.
8. Company drawing standards and templates are applied.

9. The drawing is laid out and rendered.
10. Automated validation checks completeness, conflicts and readability.
11. Low-confidence or unresolved items are flagged.
12. Engineer reviews and edits the generated drawing.
13. Approved output is exported as DWG, DXF, PDF or required native format.
14. Engineer corrections are captured as feedback for future rule/model improvements.

9. Core Capabilities

Capability	What it should accomplish
Model understanding	Extract exact geometry, topology and measurable properties.
Feature recognition	Identify manufacturing-relevant features and relationships.
View planning	Choose appropriate primary, projected, section and detail views.
Dimensioning	Generate overall, feature, chain, baseline, ordinate and coordinate dimensions where appropriate.
Hole documentation	Recognize hole patterns and generate callouts or tables.
Annotation	Generate notes, labels, symbols and other drawing information.
Tolerancing	Support general and feature-level tolerances and, eventually, GD&T; workflows.
Template management	Apply customer/company-specific standards and title blocks.
Validation	Detect missing, duplicated, conflicting or poorly placed information.
Human review	Allow engineers to edit, approve and release drawings.
Learning loop	Use corrections and approvals to improve rules and company-specific behavior.

10. Engineer-in-the-Loop

The product should automate repetitive engineering work rather than remove engineering accountability. The engineer remains the final authority over the released drawing.

Generated drawing → automated checks → flagged items → engineer review → correction → approval → release.

Corrections are valuable product data. If engineers repeatedly change a generated decision, the system can learn the company's preferred approach through updated rules, configuration or model training.

11. Recommended MVP

The first version should focus on a constrained class of parts and a limited set of drawing requirements. The objective is to prove the core pipeline rather than support every possible manufacturing scenario.

- Input: STEP files.

- Geometry extraction using an OpenCascade-based CAD kernel.
- Recognition of planes, cylinders, holes and basic feature relationships.
- Automatic front/top/right/isometric views.
- Overall dimensions and selected feature dimensions.
- Basic hole callouts and hole tables.
- One configurable company drawing template.
- PDF and DXF output.
- Basic drawing-quality checks.
- Engineer review and correction workflow.

Later phases: SolidWorks/Inventor integration, complex feature recognition, section/detail automation, advanced tolerancing and GD&T;, batch processing, additional templates, more CAD formats and deeper learning from engineering corrections.

12. Initial Technology Direction

Area	Potential technology direction
CAD geometry	OpenCascade / OpenCascade-based tooling
Backend	Python + FastAPI
Engineering logic	Python/C++ algorithms and deterministic rules
AI	Open-source LLM/VLM models where they add value; model choice should be validated against engineering tasks
Drawing generation	CAD APIs / geometry kernel plus DXF/SVG/PDF tooling
DXF	ezdxf or equivalent
Database	PostgreSQL
Job processing	Redis + Celery or equivalent queue
Frontend	React
Deployment	Docker; cloud CPU/GPU workers as required

13. Long-Term Vision

The end-state is a system where an engineer can provide a 3D model and receive a highly complete manufacturing drawing with minimal manual drafting effort.

The platform should become increasingly aware of a company's engineering practices: how it chooses views, how it dimensions recurring features, how it formats notes, how it uses tolerances, and how its engineers review drawings.

Long-term vision: DraftGPT becomes an intelligent engineering documentation layer between CAD design and manufacturing execution.

14. Product Positioning

Do not position DraftGPT as: “an AI that converts 3D models into 2D drawings.”

Position DraftGPT as: “an AI-powered manufacturing detailing platform that transforms 3D CAD designs into company-standard, validated, shop-floor-ready engineering drawings.”

The fundamental product principle is simple:

3D design is the input. Manufacturing-ready documentation is the output.